

QUESTION PAPER

Name of the Examination: CAT-1 (WIN 2023-2024)

Course Code: CSE4005

Course Title: Data Warehousing and Data Mining

Date of Exam: 09/02/2024

SET NO: 2

Duration: 90 min

(FN) (E1)

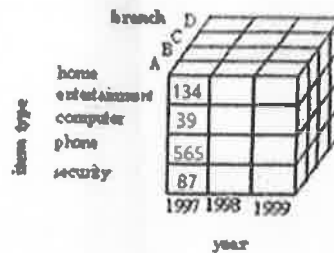
Total Marks: 50

Q1. (a) What are the differences between the ROLAP and MOLAP?

(7M)

(b) Explain OLAP operations 1. Drill Up, 2. Drill Down, 3. Slice, 4. Dice, 5. Pivot by considering the below example.

(8M)



Q2. Let us consider data for analysis includes the attribute mark. The mark values for the data tuples are (in increasing order) 23, 25, 26, 26, 29, 30, 30, 31, 32, 32, 35, 35, 35, 35, 40, 43, 43, 45, 45, 45, 45, 46, 50, 55, 56, 62, 80.

(a) What is the mean of the data?

(b) What is the median?

(c) What is the mode of the data? Comment on the data's modality (i.e., bimodal, trimodal, etc.).

(d) Can you find (roughly) the first quartile (Q1) and the third quartile (Q3) of the data?

(e) Give the five-number summary of the data. (f) Show a boxplot of the data.

(10M)

Q3. Perform the principle of component analysis for the dimensionality reduction on given "Studentmark_details" data set and Plot the PCA graph.

(10M)

Features	CAT1	CAT2	CAT3
First Year	96	88	78
Second Year	56	78	89

Q4. Describe each of the following data mining tasks and its types: predictive and descriptive. Discuss each data mining task using a real-time example.

(15M)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	CO1	PO1	1	1	15
Q2	1	CO1	PO1	1	1	10
Q3	1	CO1	PO1	1	1	10
Q4	2	CO2	PO1	1	1	15

QUESTION PAPER

Name of the Examination: CAT-1 (WIN 2023-2024)

Course Code: CSE4005

Course Title: Data Warehousing and Data Mining

Date of Exam: 09/02/2024 **SET No: 3**

Duration: 90 min (Ans) (E2) **Total Marks: 50**

Q1. (a). Compare and contrast OLAP and OLTP (7M)

(b). In real-world datasets, instances with missing values in certain attributes are frequently encountered. Discuss various approaches for addressing this issue. (8M)

Q2. Suppose a group of 12 patient medical records has been sorted as follows: 15, 20, 21, 23, 25, 45, 60, 65, 82, 102, 304, 315 Partition them into three bins by each of the following methods: (a) equal-frequency (equal depth) partitioning (b) equal-width partitioning (10M)

Q3. Find coefficient of correlation for the following dataset using Pearson's correlation. (10M)

X	Y
3	3
5	5
4	4
6	2
2	6

Q4. Describe major issues in data mining and explain the architecture of data mining with neat sketch. (15M)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	CO1	PO1	1	1	15
Q2	1	CO1	PO1	1	1	10
Q3	1	CO1	PO1	1	1	10
Q4	2	CO2	PO1	1	1	15

QUESTION PAPER

Name of the Examination: CAT-1 (WIN 2023-2024)

Course Code: CSE4005

Course Title: Data Warehousing and Data Mining

Date of Exam: 08/02/2024

SET NO: 4

Duration: 90 min (AN) (D2)

Total Marks: 50

Q1. A data warehouse can be modeled by either a star schema or a snowflake schema. Briefly describe the similarities and the differences of the two models, and then analyze their advantages and disadvantages with regard to one another. Give your opinion of which might be more empirically useful and state the reasons behind your answer **(15M)**

Q2. Using the data for age given in Table, answer the following:

Age	35	41	11	53	44	23	21	78	90
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(a) Use min-max normalization to transform the value 41 for age onto the range [0.0,1.0].

(b) Use z-score normalization to transform the value 35 for age, where the standard deviation of age is 12.94 years.

(c) Use normalization by decimal scaling to transform the value 78 for age. **(10M)**

Q3. Explain about three tier Data warehouse architecture with neat diagram. **(10M)**

Q4. Describe classification of data mining systems. **(15M)**

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	CO1	PO1	1	1	15
Q2	1	CO1	PO1	1	1	10
Q3	1	CO1	PO1	1	1	10
Q4	2	CO2	PO1	1	1	15

QUESTION PAPER

Name of the Examination: CAT-1 (WIN 2023-2024)

Course Code: CSE4005

Course Title: Data Warehousing and Data Mining

Date of Exam: 08/02/2024

SFT NO: 5

Duration: 90 min

(FN) (DI)

Total Marks: 50

Answer all the questions

Q1. Differentiate between i) OLTP and OLAP ii) Enterprise warehouse, Data Mart and Virtual Warehouse (5M+5M=10M)

Q2. Create a star schema to store data for analysis with one fact(Sales) and four dimensions (Product, Time, Customer, Employee). The fact has measures like quantity, amount, discount, and payment.

Product has Prod_id, Prod_Name, Brand, Category, price and discount.

Time has time_id, day, day_of_week, month, quarter, year.

Customer his defined by Cust_id, Cust_Name, Email_id, Mobile_No, Address

Employee has Emp_id, Emp_Name, Mobile_No, Email_id, Salary, Join Date, Last_working_date (10M)

Q3. The salary of the employees are given as follows: 10000,8000,5000,20000,15000,35000

Perform i) Min-Max Normalization in the range [0,1]

(4M)

ii) Z-score normalization

(4M)

iii) Decimal Scaling

(2M)

Q4. Compute Pearson's coefficient of correlation between advertisement cost and sales as per the data given below and what do you infer

Advertisement Cost in 1000's 39 65 62 90 82 75 25

Sales in lakhs 47 53 58 86 62 68 60.

(10M)

Q5. Explain the difference and similarity between

(10M)

i)Discrimination and Classification ii) Characterization and Clustering

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	CO1	PO1	1	1	10
Q2	1	CO1	PO1	1	1	10
Q3	1	CO1	PO1	1	1	10
Q4	2	CO2	PO1	1	1	10
Q5	2	CO2	PO1	1	1	10